

Data visualization

Data visualization is the graphical representation of data and information. It involves the use of visual elements such as charts, graphs, and maps to communicate insights and patterns hidden within the data.

Important to know

The primary goal of data visualization is to make complex datasets more understandable, accessible, and interpretable to facilitate decision-making, analysis, and storytelling. By transforming raw data into visual formats, data visualization enables users to identify trends, correlations, outliers, and other significant patterns that might not be apparent from the raw data alone. Additionally, it enhances data exploration, fosters data-driven insights, and facilitates effective communication of findings to a wide audience.

- It is the representation of the data in the graphical form.
- It allows users to use and understand patterns trends and insights of data.

Why data visualization is important?

1. Summarize.
2. Easy to understand.
3. Present data in an easy way.
4. Trends (stock market).

4 important insights from the visualization of the dataset that are as follows:

- **Quick insight form data** (*Data visualization allows for rapid comprehension of key insights from complex datasets. For example, a line graph showing sales trends over time enables stakeholders to quickly identify periods of growth or decline, helping them understand the overall performance of a product or service briefly.*)
- **Informed decision making** (*By presenting data in visual formats, decision-makers can better understand the implications of various choices. For instance, a heat map illustrating customer demographics and purchasing behavior can help a marketing team decide where to allocate resources for targeted advertising campaigns, leading to more effective and strategic decisions.*)
- **Engagement** (*Interactive and visually appealing visualizations capture the audience's attention and encourage active engagement with the data. For instance, an interactive map displaying real-time traffic patterns in a city not only informs commuters but also engages them in exploring alternative routes, thereby enhancing their overall experience and interaction with the data.*)

- **Data exploration** (*Visualization tools facilitate the exploration of datasets from different angles, enabling users to uncover hidden patterns and correlations. For example, a scatter plot displaying the relationship between a company's advertising expenditure and its revenue allows analysts to explore potential causal relationships and identify outliers, leading to deeper insights and further investigation into the underlying factors influencing business performance.*)

Key Principles of visualization

- ✓ **Simplicity:** Less is more. Avoid clutter and focus on the essentials.
- ✓ **Consistency:** Use consistent colors, fonts, and symbols to avoid confusion.
- ✓ **Accuracy:** Ensure the visual accurately represents the data.
- ✓ **Interactivity:** Modern tools allow for interactive visuals, enhancing user engagement and understanding.

Popular Data Visualization Tools:

- **Tableau:** A powerful tool offering a wide range of visualization options.
- **Power BI:** Microsoft's solution for data visualization and business intelligence.
- **D3.js:** A JavaScript library for creating custom, dynamic visuals.
- **Python Libraries:** Libraries like Matplotlib, Seaborn, and Plotly are popular among data scientists.
- **R Programming:** ggplot2, plotly.

TYPES OF PLOTS

Bar Chart:

- Explanation: A graphical representation of data using bars of varying heights or lengths.
- Example: Comparing the sales of different products in a month.

Pie Chart:

- Explanation: A circular chart divided into slices to illustrate numerical proportions.
- Example: Showing the market share of various smartphone brands.

Histogram:

- Explanation: A representation of the distribution of a dataset, like a bar chart but for frequency distribution.
- Example: Displaying the age distribution of employees in a company.

Scatter Plot:

- Explanation: A graph with points plotted to show the relationship between two sets of data.
- Example: Comparing advertising spend with sales revenue.

Line Chart:

- Explanation: A chart that displays information as a series of data points connected by straight line segments.
- Example: Tracking stock market prices over a week.

Heat Map:

- Explanation: A data visualization technique where values in a matrix are represented as colors.
- Example: Showing website activity, where darker colors represent more clicks.

Box Plot (or Whisker Plot):

- Explanation: A standardized way of displaying the dataset based on a five-number summary: minimum, first quartile, median, third quartile, and maximum.
- Example: Comparing exam scores of different classes.

Area Chart:

- Explanation: Like a line chart, but the area between the axis and the line is filled with color or shading.
- Example: Displaying the total revenue of a company over several years.

Radar (or Spider) Chart:

- Explanation: A chart that displays multivariate data on axes starting from the same point.
- Example: Comparing the features of different products.

Tree map:

- Explanation: A visualization of hierarchical data using nested rectangles.
- Example: Showing storage used by different types of files on a computer.

Geographical Map:

- Explanation: A map that displays data based on geographical areas or locations.
- Example: Highlighting areas with high crime rates in a city.

Time Series:

- Explanation: A sequence of data points indexed in time order.
- Example: Analyzing the daily temperature of a place over a year.

Violin Plot:

- Explanation: A combination of a box plot and a kernel density plot, displaying the distribution of data across different categories.
- Example: Comparing the distribution of salaries across different job titles in a company.

Bubble Chart:

- Explanation: A type of scatter plot where the data points are represented as bubbles, with the size of each bubble encoding a third variable.
- Example: Visualizing the relationship between GDP, population, and carbon emissions for different countries.

Waterfall Chart:

- Explanation: A type of floating column chart used to show how an initial value is affected by a series of positive and negative values.
- Example: Illustrating changes in the net profit of a business over successive quarters, considering factors like revenue, expenses, and taxes.

Choropleth Map:

- Explanation: A thematic map in which areas are shaded or patterned in proportion to the value of a variable being represented.
- Example: Depicting population density across different states or countries using color gradients.

Sankey Diagram:

- Explanation: A flow diagram that visualizes the magnitude of flow between nodes in a network.

- Example: Mapping energy flow in an ecosystem, illustrating how energy is transferred from producers to consumers.

Gantt Chart:

- Explanation: A horizontal bar chart used in project management to illustrate project schedules, timelines, and dependencies.
- Example: Planning and tracking the tasks and milestones of a construction project over time.

Word Cloud:

- Explanation: A visual representation of text data where the size of each word indicates its frequency or importance.
- Example: Analyzing customer feedback by creating a word cloud from survey responses to identify common themes or keywords.

Network Graph:

- Explanation: A visualization of interconnected entities represented as nodes (vertices) and edges (connections).
- Example: Visualizing social networks, where nodes represent individuals and edges represent relationships or interactions between them.

Pareto Chart:

- Explanation: A combination of a bar chart and a line graph, where bars are arranged in descending order of frequency or magnitude, and a cumulative percentage line is overlaid to show the cumulative total.
- Example: Analyzing the most significant factors contributing to product defects in a manufacturing process and identifying the "vital few" issues that account for most problems.

Sunburst Chart:

- Explanation: A hierarchical chart resembles a pie chart, where each ring represents a level in the hierarchy, and segments within the rings represent categories or subcategories.
- Example: Visualizing the organizational structure of a company, with each ring representing different levels such as departments, teams, and individual employees.

Bullet Graph:

- Explanation: A variation of a bar chart designed to compare a single, primary measure to one or more other measures, such as benchmarks or targets.
- Example: Assessing the performance of employees against key performance indicators (KPIs) by comparing their actual performance to predefined targets or benchmarks.

Hexbin Plot:

- Explanation: A two-dimensional histogram where the data points are aggregated into hexagonal bins, useful for visualizing the distribution of large datasets.
- Example: Analyzing the spatial distribution of crime incidents across a city, with each hexagon representing a region and the color intensity indicating the frequency of incidents.

Donut Chart:

- Explanation: Like a pie chart but with a hole in the center, making it easier to compare the relative sizes of categories.
- Example: Representing the distribution of expenses in a budget, with each segment of the donut chart showing the percentage of spending in different categories such as housing, transportation, and entertainment.

Error Bar Plot:

- Explanation: A graphical representation of the variability of data, typically used to show confidence intervals or standard deviations around data points.
- Example: Showing the average test scores of students in different classes along with error bars representing the standard deviation to illustrate the variability in scores.

Swarm Plot:

- Explanation: A categorical scatter plot where data points are adjusted along the categorical axis to avoid overlapping, providing a clear view of individual data points.
- Example: Visualizing the distribution of exam scores among students in different grades, with each data point representing a student's score.

Violin Swarm Plot:

- Explanation: Combines aspects of a violin plot and a swarm plot, displaying both the distribution of data as a violin plot and individual data points as a swarm plot.
- Example: Comparing the distribution of heights among different age groups, where the violin plot shows the overall distribution, and the swarm plot displays individual heights within each group.